

## ASCII MASTER FLOW – PANEL 1/12: Coastal system map

RIVER AND SEDIMENT -> material reaches the coast

WAVES TIDES CURRENTS -> transport and reshape shoreline

ECOSYSTEMS -> trap sediment buffer hazards support habitat

HUMAN USE -> extraction structures ports settlements and livelihoods

OUTCOME -> coupled land-sea response, not one isolated driver

MUST REMEMBER: Coastal and marine ecology links land-sea nutrient and sediment flows,...

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## ASCII MASTER FLOW – PANEL 2/12: Coastal ecosystem matrix

MANGROVE -> intertidal woody vegetation

SEAGRASS -> submerged flowering-plant meadow

SALT MARSH -> vegetated intertidal wetland

CORAL REEF -> biogenic marine structure

DUNE MUDFLAT ESTUARY -> distinct sediment and habitat systems

## ASCII MASTER FLOW – PANEL 3/12: Coral dual-stress diagram

THERMAL STRESS -> symbiont loss and bleaching

ACIDIFICATION -> carbonate chemistry changes

LOCAL POLLUTION -> compounding stress

RECOVERY -> event severity and local condition matter

RULE -> bleaching and acidification are not synonyms

## ASCII MASTER FLOW – PANEL 4/12: Sediment-budget rail

SOURCE -> rivers cliffs reefs and alongshore supply

TRANSPORT -> waves currents and tides

INTERRUPTION -> dams mining ports and coastal structures

EVENT -> storm and surge redistribution

SHORELINE -> erosion accretion or reorientation

## ASCII MASTER FLOW – PANEL 5/12: CRZ legal hierarchy

ENVIRONMENT PROTECTION ACT -> enabling legal base

CRZ NOTIFICATION -> national regulatory framework

AMENDMENT -> changes specified provisions

CZMP -> mapped state or UT implementation layer

PROJECT DECISION -> separate appraisal and clearance

## ASCII MASTER FLOW – PANEL 6/12: CRZ category compass

CRZ I -> ecologically sensitive and intertidal contexts  
CRZ II -> developed urban coastal context  
CRZ III -> relatively undisturbed rural context  
CRZ IV -> specified water-area context  
SUBCATEGORY RULE -> official notification controls detail

## ASCII MASTER FLOW – PANEL 7/12: Coastal line firewall

HTL -> mapped tidal reference

LTL -> mapped lower tidal reference

HAZARD LINE -> hazard-information function

SETBACK OR NDZ -> rule-defined regulatory function

APPROVED MAP -> category boundary evidence

## ASCII MASTER FLOW – PANEL 8/12: CZMP-to-clearance sequence

CZMP -> category and boundary map

PROJECT LOCATION -> overlaid on approved plan

APPLICABLE RULE -> category subcategory and activity

APPRAISAL OR CLEARANCE -> project-specific decision

COMPLIANCE -> construction and operation monitoring

## ASCII MASTER FLOW – PANEL 9/12: ICZM coordination wheel

ECOSYSTEMS -> cumulative ecological condition

HAZARDS -> erosion surge flooding and intrusion

LIVELIHOODS -> fishing access tenure and settlements

DEVELOPMENT -> ports tourism energy and infrastructure

COORDINATION -> wider than one project clearance

## ASCII MASTER FLOW – PANEL 10/12: Coastal aquifer mechanism

OVER EXTRACTION OR LOWER RECHARGE -> freshwater head falls

HYDRAULIC GRADIENT CHANGES -> saline interface moves

SEA LEVEL OR ENGINEERING -> possible compounding pressure

SALINISATION -> wells soils and supply affected

RESPONSE -> demand recharge pumping monitoring and barriers

CLOSE DISTINCTION: CRZ category is not protected-area category, HTL is not an arbitrary...

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## ASCII MASTER FLOW – PANEL 11/12: Blue Economy sustainability test

SECTOR OPPORTUNITY -> fisheries shipping tourism energy or science

ECOSYSTEM PRESSURE -> habitat pollution extraction and carbon

LIVELIHOOD AND EQUITY -> access benefit and displacement

CUMULATIVE GOVERNANCE -> thresholds monitoring and enforcement

VERDICT -> growth is not sustainability without outcomes

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State CRZ notification and amendment...

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## ASCII MASTER FLOW – PANEL 12/12: Coastal answer spine

DEFINE -> ecological coast and administrative CRZ separately

MAP -> habitat sediment hazard aquifer and livelihood systems

PLACE -> notification CZMP category clearance ICZM and maritime zone

EVALUATE -> sector opportunity against ecosystem and equity outcome

VERIFY -> official map survey notification decision and date